8.2 Weighted Average of Points (Section Formula)

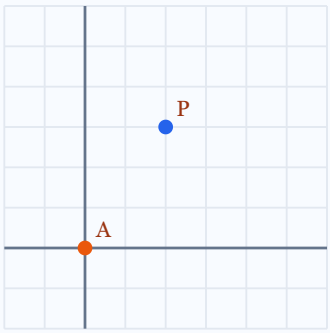
A point P that divides $\overline{AB}$ in the ratio $AP : PB = m : n$ is a **weighted average** of the endpoints — it sits closer to whichever endpoint has the larger share of the ratio. Since P is a fraction $\frac{m}{m+n}$ of the way from A to B, you can find it with $P = A + \frac{m}{m+n}(B - A)$, which expands to the section formula $P = \left(\frac{nx_A + mx_B}{m+n}, \frac{ny_A + my_B}{m+n}\right)$. When $m = n$ (ratio $1 : 1$), this collapses to the ordinary midpoint formula — the midpoint is just the special case of a $1 : 1$ weighted average. This worksheet also runs the formula in reverse: given the partition point and one endpoint, find the missing endpoint.

Examples

Example 1: Ratio $1:2$ from A

$A(0, 0), B(6, 9)$. Point P divides $\overline{AB}$ so that $AP : PB = 1 : 2$. Find P.

Solution



P is $\frac{1}{3}$ of the way from A to B.

P is $\frac{m}{m+n} = \frac{1}{3}$ of the way from A to B:

$P = A + \frac{1}{3}(B - A) = (0, 0) + \frac{1}{3}(6, 9) = (2, 3).$

Answer: $P = (2, 3)$.

Example 2: Ratio $2:1$ from A

$A(1,2), B(7,8)$. Point P divides $\overline{AB}$ so that $AP : PB = 2 : 1$. Find P.

Solution

Use the section formula with $m = 2, n = 1$:

$$x_P = \frac{n \cdot x_A + m \cdot x_B}{m+n} = \frac{1(1)+2(7)}{3} = \frac{15}{3} = 5.$$

$$y_P = \frac{1(2)+2(8)}{3} = \frac{18}{3} = 6.$$

Answer: $P = (5,6)$.

Example 3: Midpoint as the $1:1$ special case

$A(4,-2), B(10,6)$. Find the point that divides $\overline{AB}$ in ratio $1 : 1$.

Solution

With $m = n = 1$, the section formula becomes $P = (\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2})$ — the ordinary midpoint formula.

$$P = (\frac{4+10}{2}, \frac{-2+6}{2}) = (7,2).$$

Answer: $P = (7,2)$ (the midpoint).

Example 4: Fraction form

Find the point that is $\frac{3}{4}$ of the way from $A(0,0)$ to $B(8,4)$.

Solution

Identify the endpoints and the fraction of the way from A to B: $A = (0,0)$, $B = (8,4)$, fraction $= \frac{3}{4}$.

Apply $P = A + \frac{3}{4}(B - A)$:

$$P = A + \frac{3}{4}(B-A) = (0,0) + \frac{3}{4}(8,4) = (6,3).$$

This point sits three-quarters of the way from A toward B.

Answer: $(6,3)$.

Example 5: Where the formula comes from

Explain why $P = A + \frac{m}{m+n}(B - A)$ is equivalent to $P = (\frac{nx_A + mx_B}{m+n}, \frac{ny_A + my_B}{m+n})$.

Solution

Start with the x-coordinate: $x_P = x_A + \frac{m}{m+n}(x_B - x_A)$.

Distribute: $x_P = x_A + \frac{m\,x_B}{m+n} - \frac{m\,x_A}{m+n} = \frac{(m+n)x_A - m\,x_A + m\,x_B}{m+n} = \frac{n\,x_A + m\,x_B}{m+n}$.

The same algebra applies to y. Both forms give the same point — the second form is just easier to compute directly.

Example 6: Ratio $3:1$

$A(-2, 5), B(6, 1); AP : PB = 3 : 1$. Find P.

Solution

$m = 3, n = 1$:

$x_P = \frac{1(-2) + 3(6)}{4} = \frac{-2 + 18}{4} = \frac{16}{4} = 4.$

$y_P = \frac{1(5) + 3(1)}{4} = \frac{5 + 3}{4} = \frac{8}{4} = 2.$

Answer: $P = (4, 2)$.

Example 7: One-fourth of the way

Find the point $\frac{1}{4}$ of the way from $(2, 3)$ to $(10, 11)$.

Solution

Identify the endpoints and the fraction: starting point $(2, 3)$, ending point $(10, 11)$, fraction $= \frac{1}{4}$.

Apply the weighted-average formula:

$P = (2, 3) + \frac{1}{4}((10, 11) - (2, 3)) = (2, 3) + \frac{1}{4}(8, 8) = (2, 3) + (2, 2) = (4, 5).$

This point lies one-fourth of the way along the segment from $(2, 3)$ to $(10, 11)$.

Answer: $(4, 5)$.

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Example 8: Ratio to fraction

For ratio $AP : PB = 2 : 3$, what fraction of the way from A to B is P?

Solution

The fraction of the way from A is always $\frac{m}{m+n}$ where $m : n$ is the ratio from A: $\frac{2}{2+3} = \frac{2}{5}$.

Answer: $\frac{2}{5}$.

Example 9: Partition giving a decimal

$A(0, 8), B(10, 0)$; ratio $1 : 4$ from A. Find P.

Solution

Identify m and n from the ratio $AP : PB = 1 : 4$, so $m = 1, n = 4$.

Apply the section formula:

$$x_P = \frac{4(0) + 1(10)}{5} = \frac{10}{5} = 2.$$

$$y_P = \frac{4(8) + 1(0)}{5} = \frac{32}{5} = 6.4.$$

The point need not have whole-number coordinates.

Answer: $P = (2, 6.4)$.

Example 10: Work backward to find the missing endpoint

Point $P(2, 3)$ divides $\overline{AB}$ so that $AP : PB = 1 : 3$, and $A = (0, 0)$. Find B.

Solution

The fraction of the way from A to B is $\frac{m}{m+n} = \frac{1}{4}$, so $P = A + \frac{1}{4}(B - A)$.

Since $A = (0, 0)$, this simplifies to $P = \frac{1}{4}B$.

Solve for B: $B = 4P = 4(2, 3) = (8, 12)$.

Answer: $B = (8, 12)$.

Example 11: Application — rest stop on a trail

A hiking trail runs from the trailhead $T(0, 0)$ to the summit $S(20, 15)$ (coordinates in hundreds of meters). A rest stop sits exactly $\frac{2}{5}$ of the way from T to S. Find the coordinates of the rest stop.

Solution

$$P = T + \frac{2}{5}(S - T) = (0, 0) + \frac{2}{5}(20, 15) = (\frac{2}{5} \cdot 20, \frac{2}{5} \cdot 15) = (8, 6).$$

Answer: the rest stop is at $(8, 6)$, i.e., 800 m east and 600 m north of the trailhead.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Ratio $1:1$

$A(2, 4), B(8, 10)$. Midpoint?

Answer: **Verification:**

Problem 2: Ratio $1:2$

$A(0, 0), B(9, 6); AP : PB = 1 : 2$. Find P.

Answer: **Verification:**

Problem 3: Ratio $2:1$

$A(1, 1), B(4, 7); AP : PB = 2 : 1$. Find P.

Answer: **Verification:**

Problem 4: Fraction

$\frac{1}{4}$ of the way from $(0, 0)$ to $(12, 8)$.

Answer: **Verification:**

Problem 5: Fraction

$\frac{3}{4}$ of the way from $(0, 0)$ to $(12, 8)$.

Answer: **Verification:**

Problem 6: Ratio $3:2$

$A(0, 0), B(10, 15); AP : PB = 3 : 2$. Find P.

Answer: **Verification:**

Problem 7: Fraction

What fraction of the way is ratio $1:3$ from A?

Answer: **Verification:**

Problem 8: Partition

$A(-4,2), B(8,8); AP:PB = 1:2$. Find P.

Answer: **Verification:**

Problem 9: Ratio $3:1$

$A(2,0), B(6,8); AP:PB = 3:1$. Find P.

Answer: **Verification:**

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Problem 10: Midpoint

Ratio $1:1$ gives the...

Answer: **Verification:**

Problem 11: Partition

$A(5, 5)$, $B(5, 17)$; halfway point?

Answer: **Verification:**

Problem 12: Fraction

$\frac{2}{5}$ of the way from $(1, 2)$ to $(11, 12)$.

Answer: **Verification:**

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Problem 13: Ratio $2:3$

$A(0,6), B(10,1); AP : PB = 2 : 3$. Find P.

Answer: **Verification:**

Problem 14: Challenge

Point $P(4,5)$ divides $\overline{AB}$ so that $AP : PB = 3 : 5$, with $A(1,2)$. Find B.

Answer: **Verification:**

Problem 15: Challenge

$A(2,-1), B(14,11); AP : PB = 3 : 1$. Find P, then verify $AP + PB = AB$ using the distance formula.

Answer: **Verification:**

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✓ Answer Key

1. $(5, 7)$
2. $(3, 2)$
3. $(3, 5)$
4. $(3, 2)$
5. $(9, 6)$
6. $(6, 9)$
7. $\frac{1}{4}$
8. $(0, 4)$
9. $(5, 6)$
10. Midpoint
11. $(5, 11)$
12. $(5, 6)$
13. $(4, 4)$
14. $B = (9, 10)$
15. $P = (11, 8)$; $AP = 9\sqrt{2}$, $PB = 3\sqrt{2}$, $AB = 12\sqrt{2}$, and $9\sqrt{2} + 3\sqrt{2} = 12\sqrt{2}$

Great work — keep practising!